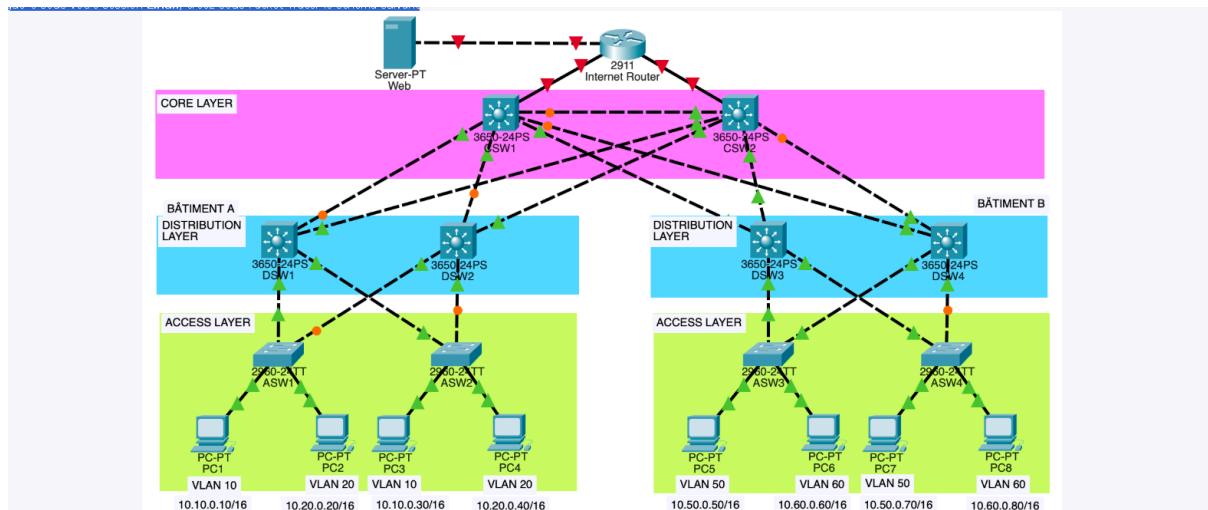
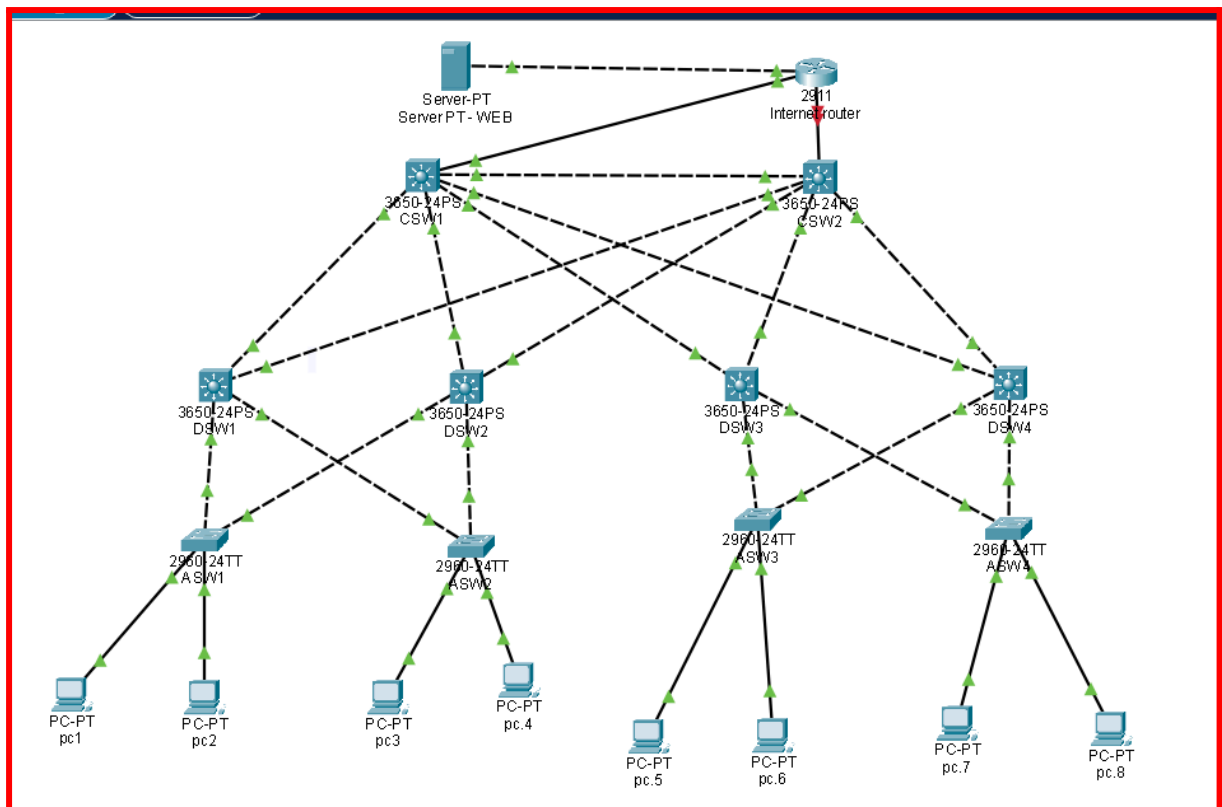


Réseaux de Campus (CAN) et Architecture à 3 couches (3-Tier Architecture)

Travail demandé

- Logué-e sous votre session **Linux**, créez sous Packet Tracer le schéma suivant





- Configurez les PC avec les configurations suivantes.

PC	VLAN	Adresse	Passerelle
PC1	VLAN 10	10.10.0.10/16	10.10.0.1/16
PC3		10.10.0.30/16	
PC2	VLAN 20	10.20.0.20/16	10.20.0.1/16
PC4		10.20.0.40/16	
PC5	VLAN 50	10.50.0.50/16	10.50.0.1/16
PC7		10.50.0.70/16	
PC6	VLAN 60	10.60.0.60/16	10.60.0.1/16
PC8		10.60.0.80/16	

- Configurez les switches d'accès (VLAN et Trunks).
- Créez les VLAN sur les switches de distribution.
- Vérifiez que les PC d'un même VLAN se pingent.
 - Et les PC de VLAN différents ?
 - réponse : non... Il va falloir ajouter du routage pour cela.

ASW1

- enable
- configure terminal
- vlan 10
- name VLAN10
- exit

- vlan 20
- name VLAN20
- exit
-
- interface fastEthernet0/3
- switchport mode access
- switchport access vlan 10
- no shutdown
- exit
-
- interface fastEthernet0/4
- switchport mode access
- switchport access vlan 20
- no shutdown
- exit
-
- ! Trunk link to DSW1
- interface fastEthernet0/1
- switchport mode trunk
- switchport trunk allowed vlan 10,20
- switchport trunk native vlan 1
- no shutdown
- exit
-
- ! Trunk link to DSW2
- interface fastEthernet0/2
- switchport mode trunk
- switchport trunk allowed vlan 10,20
- switchport trunk native vlan 1
- no shutdown
- exit
-
- end
- write memory

ASW2

- enable
- configure terminal
-
- ! Create VLANs
- vlan 10
- name VLAN10
- exit
- vlan 20
- name VLAN20
- exit
-
- ! Access ports for PCs

- interface fastEthernet0/3
- switchport mode access
- switchport access vlan 10
- no shutdown
- exit
-
- interface fastEthernet0/4
- switchport mode access
- switchport access vlan 20
- no shutdown
- exit
-
- ! Trunk link to DSW1
- interface fastEthernet0/1
- switchport mode trunk
- switchport trunk allowed vlan 10,20
- switchport trunk native vlan 1
- no shutdown
- exit
-
- ! Trunk link to DSW2
- interface fastEthernet0/2
- switchport mode trunk
- switchport trunk allowed vlan 10,20
- switchport trunk native vlan 1
- no shutdown
- exit
-
- end
- write memory

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	pc1	pc3	ICMP		0.000	N	0	(edit)	(delete)
	Successful	pc2	pc4	ICMP		0.000	N	1	(edit)	(delete)
	Successful	pc5	pc7	ICMP		0.000	N	2	(edit)	(delete)
	Successful	pc6	pc8	ICMP		0.000	N	3	(edit)	(delete)

DSW	VLAN	Adresse
DSW1	VLAN 10	10.10.0.100/16
	VLAN 20	10.20.0.100/16
DSW2	VLAN 10	10.10.0.200/16
	VLAN 20	10.20.0.200/16
DSW3	VLAN 50	10.50.0.100/16
	VLAN 60	10.60.0.100/16
DSW4	VLAN 50	10.50.0.200/16
	VLAN 60	10.60.0.200/16

- **DSW1 sera prioritaire à DSW2 pour la passerelle 10.10.0.1 du VLAN 10.**
- **DSW2 sera prioritaire à DSW1 pour la passerelle 10.20.0.1 du VLAN 20.**
- **DSW3 sera prioritaire à DSW4 pour la passerelle 10.50.0.1 du VLAN 50.**
- **DSW4 sera prioritaire à DSW3 pour la passerelle 10.60.0.1 du VLAN 60.**

DSW1

enable

conf t

ip routing

! VLAN10

interface vlan 10

ip address 10.10.0.100 255.255.0.0

standby 10 ip 10.10.0.1

standby 10 priority 105

standby 10 preempt

no shutdown

! VLAN20

interface vlan 20

ip address 10.20.0.100 255.255.0.0

standby 20 ip 10.20.0.1

standby 20 priority 95

standby 20 preempt

no shutdown

End

DSW2

enable

conf t

ip routing

```

! VLAN10
interface vlan 10
ip address 10.10.0.200 255.255.0.0
standby 10 ip 10.10.0.1
standby 10 priority 95
standby 10 preempt
no shutdown

```

```

! VLAN20
interface vlan 20
ip address 10.20.0.200 255.255.0.0
standby 20 ip 10.20.0.1
standby 20 priority 105
standby 20 preempt
no shutdown

```

```
end
```

```

DSW1>en
DSW1#show standby brief

```

Interface	Grp	Pri	P	State	Active	Standby	Virtual IP
Vl10	10	105	P	Active	local	10.10.0.200	10.10.0.1
Vl20	20	95	P	Standby	10.20.0.200	local	10.20.0.1

```

DSW1#

```

DSW3

```

enable
conf t
ip routing

```

```

! VLAN50
interface vlan 50
ip address 10.50.0.100 255.255.0.0
standby 50 ip 10.50.0.1
standby 50 priority 105
standby 50 preempt
no shutdown

```

```

! VLAN60
interface vlan 60
ip address 10.60.0.100 255.255.0.0
standby 60 ip 10.60.0.1
standby 60 priority 95

```

```
standby 60 preempt
no shutdown
```

```
end
```

DSW4

```
enable
conf t
ip routing
```

! VLAN50

```
interface vlan 50
ip address 10.50.0.200 255.255.0.0
standby 50 ip 10.50.0.1
standby 50 priority 95
standby 50 preempt
no shutdown
```

! VLAN60

```
interface vlan 60
ip address 10.60.0.200 255.255.0.0
standby 60 ip 10.60.0.1
standby 60 priority 105
standby 60 preempt
no shutdown
```

```
end
```

```
DSW3>en
DSW3#show standby brief
                P indicates configured to preempt.
                |
Interface      Grp  Pri  P State      Active        Standby        Virtual IP
Vl50           50   105  P Active     local         unknown       10.50.0.1
Vl60           60   95   P Active     local         unknown       10.60.0.1
DSW3#
```

- Configurez de la même manière DSW2 avec en particulier une priorité de 95 pour 10.10.0.1 (il prendra cette passerelle si DSW1 plus disponible) et une priorité de 105 pour 10.20.0.1 (il prend cette passerelle, et DSW1 sera en secours pour celle-ci).

```
DSW2#show standby brief
                P indicates configured to preempt.
                |
Interface      Grp  Pri  P State      Active        Standby        Virtual IP
Vl10           10   95   P Standby    10.10.0.100   local          10.10.0.1
Vl20           20   105 P Active     local         10.20.0.100   10.20.0.1
DSW2#
```

- Configurez les VLAN 50 et 60 sur DSW3 et DSW4.

DSW3:

enable

conf t

ip routing

! VLAN50 (DSW3 actif)

interface vlan 50

ip address 10.50.0.100 255.255.0.0

standby 50 ip 10.50.0.1

standby 50 priority 105

standby 50 preempt

no shutdown

! VLAN60 (DSW3 en secours)

interface vlan 60

ip address 10.60.0.100 255.255.0.0

standby 60 ip 10.60.0.1

standby 60 priority 95

standby 60 preempt

no shutdown

end


```

DSW3#
%SYS-5-CONFIG_I: Configured from console by console

DSW3#
DSW3#show standby brief
                P indicates configured to preempt.
                |
Interface      Grp   Pri  P State      Active        Standby        Virtual IP
Vl50           50   105 P Listen    10.50.0.100   unknown       10.50.0.1
Vl60           60   95  P Listen    10.60.0.100   unknown       10.60.0.1
DSW3#

```

DSW4

enable

conf t

ip routing

! VLAN50 (DSW4 en secours)

interface vlan 50

ip address 10.50.0.200 255.255.0.0

standby 50 ip 10.50.0.1

standby 50 priority 95

standby 50 preempt

no shutdown

! VLAN60 (DSW4 actif)

interface vlan 60

ip address 10.60.0.200 255.255.0.0

standby 60 ip 10.60.0.1

standby 60 priority 105

standby 60 preempt

no shutdown

end

```
DSW4#
DSW4#show standby brief
                P indicates configured to preempt.
                |
Interface      Grp  Pri  P State      Active        Standby        Virtual IP
V150           50   95   P Listen     10.50.0.200   unknown        10.50.0.1
V160           60   105 P Listen     10.60.0.200   unknown        10.60.0.1
DSW4#
```

THE codes

INTERNET ROUTER (2911)

- enable
- configure terminal
- hostname InternetRouter
- ip routing
-
-
- ! === Link to Core S3
- interface gig0/0
- description Link_to_S3
- ip address 10.0.5.1 255.255.255.0
- no shutdown
-
-
- ! === Link to Core S4
- interface gig0/1
- description Link_to_S4
- ip address 10.0.6.1 255.255.255.0
- no shutdown
-

-
- ! === Link to Web Server (Fa0/1/0)
- interface fa0/1/0
- description Link_to_WebServer
- ip address 10.0.10.1 255.255.255.0
- no shutdown
-
-
- exit
-
-
- ! === Static routes to internal VLANs through the core
- ip route 192.168.10.0 255.255.255.0 10.0.5.3
- ip route 192.168.20.0 255.255.255.0 10.0.5.3
-
-
- end
- write memory

S1 (Access Switch)

- enable
- configure terminal
- hostname S1
-
- vlan 10
- name VLAN10
- vlan 20
- name VLAN20
-
- interface range fa0/2-3
- switchport mode access
- switchport access vlan 10
-
- interface range fa0/4
- switchport mode access
- switchport access vlan 20
-
- interface fa0/1
- switchport trunk encapsulation dot1q

- switchport mode trunk
- no shutdown
-
- exit
- end

S2 (Multilayer Switch – Primary for VLAN 10)

- enable
- configure terminal
- hostname S2
-
- ! --- Enable Layer 3 routing
- ip routing
-
- ! --- Create VLANs
- vlan 10
- name VLAN10
- vlan 20
- name VLAN20
-
- ! --- VLAN 10 (Primary)
- interface vlan 10
- ip address 192.168.10.2 255.255.255.0
- standby 10 ip 192.168.10.1
- standby 10 priority 110
- standby 10 preempt
- no shutdown
-
- ! --- VLAN 20 (Secondary)
- interface vlan 20
- ip address 192.168.20.2 255.255.255.0
- standby 20 ip 192.168.20.1
- standby 20 priority 90
- standby 20 preempt
- no shutdown
-
-
- ! --- Trunks (adjust if needed)
- interface gig1/0/1
- description Link_to_S4
- switchport mode trunk
- no shutdown
-

- interface gig1/0/2
- description Link_to_S3
- switchport mode trunk
- no shutdown
-
- interface gig1/0/3
- description Link_to_S1
- switchport mode trunk
- no shutdown
-
-
- ! --- Default route toward core
- ip route 0.0.0.0 0.0.0.0 10.0.2.3
-
- end
- write memory

S3 (Core Switch)

- enable
- configure terminal
- hostname S3
- ip routing
-
- ! --- Routed link to S2
- interface gig1/0/2
- no switchport
- description Link_to_S2
- ip address 10.0.2.3 255.255.255.0
- no shutdown
-
- ! --- Routed link to S5
- interface gig1/0/1
- no switchport
- description Link_to_S5

- ip address 10.0.3.3 255.255.255.0
- no shutdown
-
- ! --- Routed link to Internet router
- interface gig1/0/3
- no switchport
- description Link_to_InternetRouter
- ip address 10.0.5.3 255.255.255.0
- no shutdown
-
- exit
-
- ! --- Route to the web-server network
- ip route 10.0.10.0 255.255.255.0 10.0.5.1
-
- end
- write memory

```

S3#
S3#enable
S3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
S3(config)#hostname S3
S3(config)#ip routing
S3(config)#
S3(config)#! --- Routed link to S2
S3(config)#interface gig1/0/2
S3(config-if)# no switchport
S3(config-if)# description Link_to_S2
S3(config-if)# ip address 10.0.2.3 255.255.255.0
S3(config-if)# no shutdown
S3(config-if)#
S3(config-if)#! --- Routed link to S5
S3(config-if)#interface gig1/0/1
S3(config-if)# no switchport
S3(config-if)# description Link_to_S5
S3(config-if)# ip address 10.0.3.3 255.255.255.0
S3(config-if)# no shutdown
S3(config-if)#
S3(config-if)#! --- Routed link to Internet router
S3(config-if)#interface gig1/0/3
S3(config-if)# no switchport
S3(config-if)# description Link_to_InternetRouter
S3(config-if)# ip address 10.0.5.3 255.255.255.0
S3(config-if)# no shutdown
S3(config-if)#
S3(config-if)#exit
S3(config)#
S3(config)#! --- Route to the web-server network
S3(config)#ip route 10.0.10.0 255.255.255.0 10.0.5.1
S3(config)#
S3(config)#end

```

S4 (Core Switch)

- enable
- configure terminal
- hostname S4
-
- ! --- Enable Layer 3 routing
- ip routing
-
-
- ! --- Configure routed link to S2 (Distribution)
- interface gig1/0/1
- no switchport
- description Link_to_S2
- ip address 10.0.2.4 255.255.255.0
- no shutdown
-
-
- ! --- Configure routed link to S5 (Distribution)
- interface gig1/0/2
- no switchport
- description Link_to_S5
- ip address 10.0.7.4 255.255.255.0
- no shutdown
-
-
- ! --- Configure routed link to Internet Router
- interface gig1/0/3
- no switchport
- description Link_to_InternetRouter
- ip address 10.0.6.4 255.255.255.0
- no shutdown
-
-
- exit
-

- ! --- Static route to server network via Internet router
- ip route 10.0.10.0 255.255.255.0 10.0.6.1
-
-
- end
- write memory

S5 (Multilayer Switch – Primary for VLAN 20)

- enable
- configure terminal
- hostname S5
-
- ! --- Enable L3 routing
- ip routing
-
- ! --- Create VLANs
- vlan 10
- name VLAN10
- vlan 20
- name VLAN20
-
- ! --- Interface VLAN 10 (secondary)
- interface vlan 10
- ip address 192.168.10.5 255.255.255.0
- standby 10 ip 192.168.10.1
- standby 10 priority 90
- standby 10 preempt
- no shutdown
-
- ! --- Interface VLAN 20 (primary)
- interface vlan 20
- ip address 192.168.20.5 255.255.255.0
- standby 20 ip 192.168.20.1

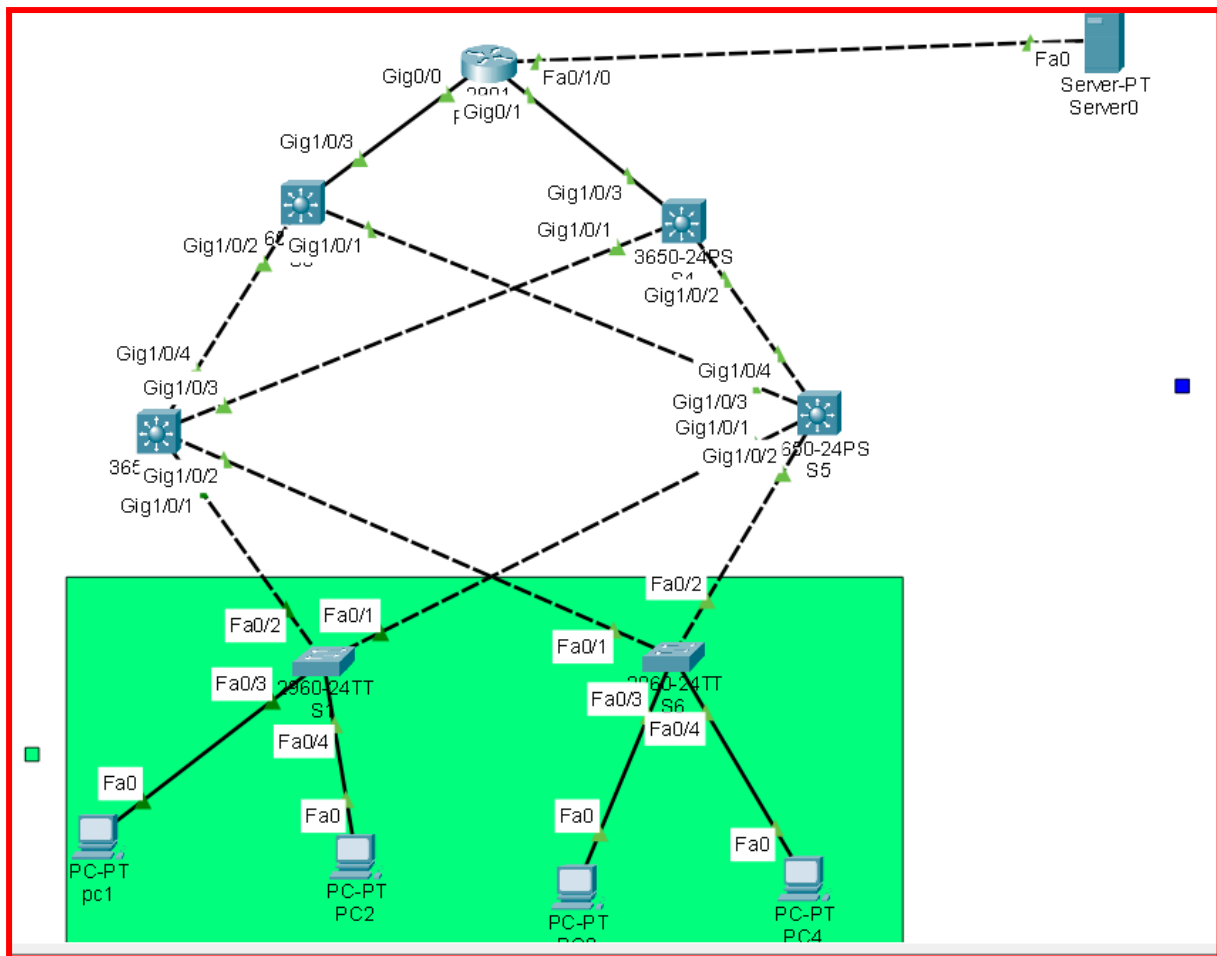
- standby 20 priority 110
- standby 20 preempt
- no shutdown
-
- ! --- Trunks (no encapsulation command)
- interface gig1/0/1
- description Link_to_S3
- switchport mode trunk
- no shutdown
-
- interface gig1/0/2
- description Link_to_S4
- switchport mode trunk
- no shutdown
-
- interface gig1/0/3
- description Link_to_S6
- switchport mode trunk
- no shutdown
-
- ! --- Optional safety tweak
- no spanning-tree portfast trunk
-
- exit
-
- ! --- Default route to core
- ip route 0.0.0.0 0.0.0.0 10.0.3.3
-
- end
- write memory

☐ **S6 (Access Switch)**

- ☐ enable
- ☐ configure terminal

- ☐ hostname S6
- ☐
- ☐ ! --- Create VLANs for user networks
- ☐ vlan 10
- ☐ name VLAN10
- ☐ vlan 20
- ☐ name VLAN20
- ☐
- ☐
- ☐ ! --- Assign access ports for PCs
- ☐ ! Adjust Fa0/x to match where your PCs are connected
- ☐ interface fa0/3
- ☐ switchport mode access
- ☐ switchport access vlan 10
- ☐ no shutdown
- ☐
- ☐ interface fa0/4
- ☐ switchport mode access
- ☐ switchport access vlan 20
- ☐ no shutdown
- ☐
- ☐
- ☐ ! --- Configure trunk link to distribution switch (S5)
- ☐ interface fa0/2
- ☐ switchport mode trunk
- ☐ no shutdown
- ☐
- ☐
- ☐ exit
- ☐
- ☐ ! --- Optional: enable portfast on access ports
- ☐ spanning-tree portfast default
- ☐
- ☐ end
- ☐ write memory

```
S6>
S6>
S6>en
S6#enable
S6#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
S6(config)#hostname S6
S6(config)#
S6(config)#! --- Create VLANs for user networks
S6(config)#vlan 10
S6(config-vlan)# name VLAN10
S6(config-vlan)#vlan 20
S6(config-vlan)# name VLAN20
S6(config-vlan)#
S6(config-vlan)#
S6(config-vlan)#! --- Assign access ports for PCs
S6(config-vlan)#! Adjust Fa0/x to match where your PCs are connected
S6(config-vlan)#interface fa0/3
S6(config-if)# switchport mode access
S6(config-if)# switchport access vlan 10
S6(config-if)# no shutdown
S6(config-if)#
S6(config-if)#interface fa0/4
S6(config-if)# switchport mode access
S6(config-if)# switchport access vlan 20
S6(config-if)# no shutdown
S6(config-if)#
S6(config-if)#
S6(config-if)#! --- Configure trunk link to distribution switch (S5)
S6(config-if)#interface fa0/2
S6(config-if)# switchport mode trunk
S6(config-if)# no shutdown
S6(config-if)#
S6(config-if)#
S6(config-if)#exit
S6(config)#
S6(config)#! --- Optional: enable portfast on access ports
S6(config)#spanning-tree portfast default
```



Server Configuration (Server-PT)

Network settings

In Packet Tracer, click the **Server**, then go to:
Desktop → IP Configuration

Enter the following:

Field	Value
IP Address	10.0.10.2
Subnet Mask	255.255.255.0
Default Gateway	10.0.10.1 (the Internet router interface)
DNS Server	<i>(optional, can leave blank or use 8.8.8.8)</i>